Assignment 1 Group 22

Question 1

```
Residuals:
     Min       1Q   Median       3Q      Max
-0.37383 -0.07047 -0.01043  0.07115  0.33309

Coefficients:
             Estimate Std. Error t value            Pr(>|t|)
(Intercept)  5.533371   0.255719   21.64 <0.0000000000000002
L1.lnsales   0.356853   0.027514   12.97 <0.0000000000000002
lngoogle.ads 0.033472   0.002263   14.79 <0.0000000000000002
lnfacebook   0.030489   0.001742   17.50 <0.0000000000000002
lntiktok     0.036573   0.001753   20.86 <0.0000000000000002

(Intercept)  ***
L1.lnsales   ***
lngoogle.ads ***
lnfacebook   ***
lntiktok     ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1024 on 194 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.8557,    Adjusted R-squared:  0.8527
F-statistic: 287.6 on 4 and 194 DF,  p-value: < 0.00000000000000022
```

Interpretation:

Formula:

$$\ln(Sales_t) = 5.533 + 0.356\ \ln(sales_{t-1}) + 0.033\ \ln(google.ads_t) + 0.030\ \ln(facebook_t) + 0.036\ \ln(tiktok_t)$$

The estimated effect size is 0.356. This implies that if today's web traffic performance gains momentum, 64.4% (1 – 0.356) of its impact carries over into the next period.

Using the log–log model, the estimated coefficients can be directly interpreted as elasticities of the variables.

The R-squared value of 85.27% indicates a strong model fit, with Google Ads, Facebook, and TikTok jointly explaining about 85% of the variation in sales.

The F-statistic (287.6, p ≈ 0.001) confirms that the model is highly significant, which means at least one predictor variable significantly explains sales.

Residuals range from –0.374 to 0.333, with a median close to zero (–0.01043), suggesting that errors

are roughly centered. The first quartile (–0.07047) and third quartile (0.07115) also indicate residuals are fairly symmetric.

Regression results indicate the following elasticities:

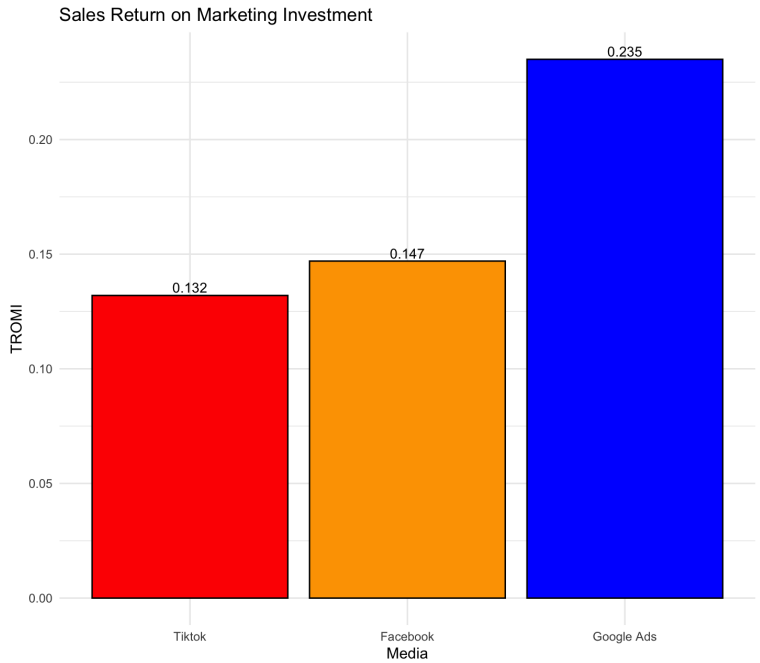
- A 1% increase in TikTok advertising is associated with a 0.037% increase in sales, holding other factors constant.
- A 1% increase in Facebook advertising is associated with a 0.030% increase in sales, holding other factors constant.
- A 1% increase in Google advertising is associated with a 0.033% increase in sales, holding other factors constant.

All coefficients are statistically significant ($p < 0.001$), indicating that each advertising channel contributes meaningfully to explaining sales performance.

Question 2

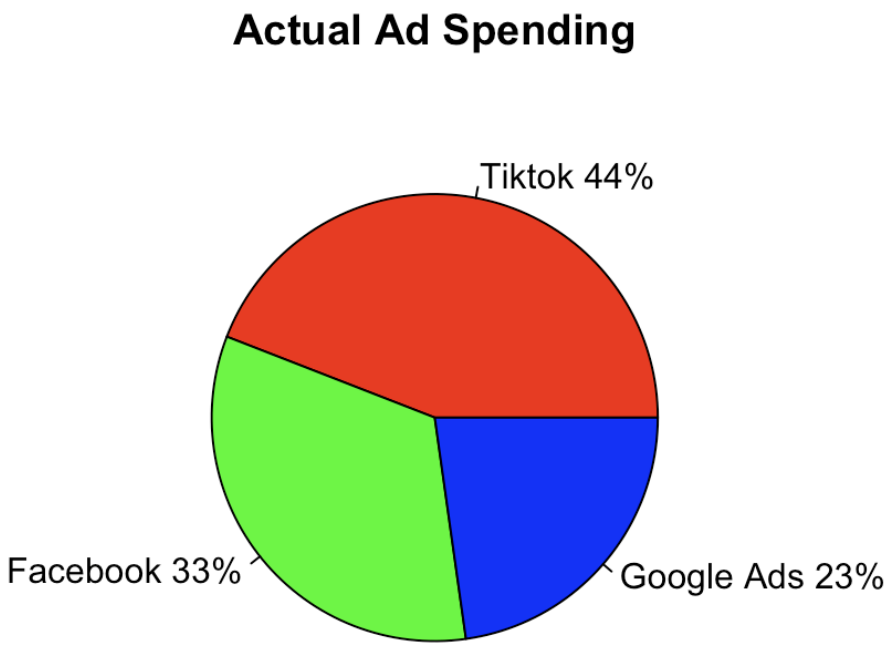
The table and bar chart below present the estimated return on investment (ROI) for each media channel. These values indicate the relative efficiency of advertising spend across TikTok, Facebook, and Google in generating sales outcomes.

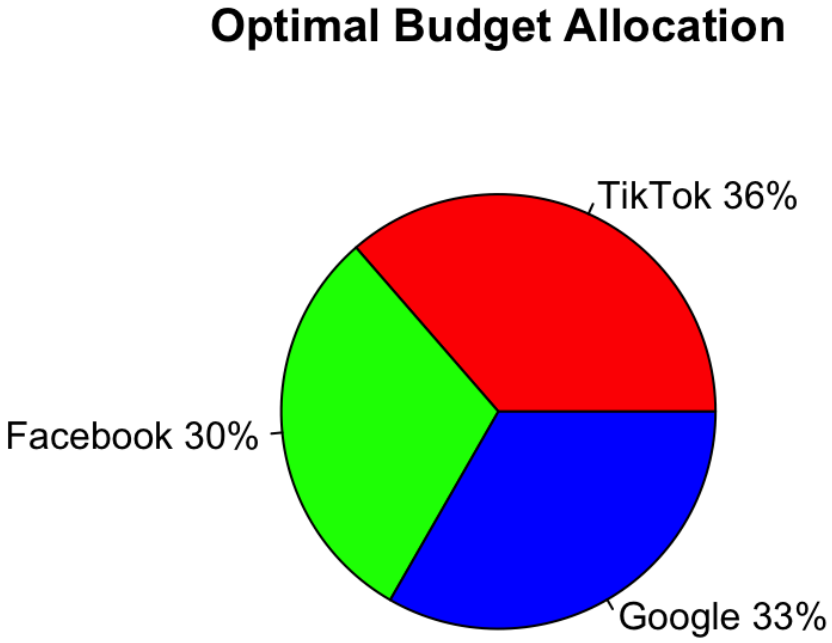
Media Names	Return On Investment
Tiktok	0.132
Facebook	0.147
Google	0.235



Compared to TikTok and Facebook, Google ads showed the highest ROI, which suggests that Google ads are the most efficient channel.

Question 3





The model-recommended allocation differs from the current spending. The pie charts show that Google Ads receives 23% of the budget, but should receive around one-third, while TikTok is currently overallocated at 44% compared to closer to 36%. Facebook also shows a slight reduction from its current share. To maximise sales, FourTex should reallocate its budget by shifting more

investment into Google Ads while reducing the share currently allocated to TikTok and Facebook. This adjustment better aligns the budget with the channels' effectiveness and ensures higher overall returns.

Question 4

```
Residuals:
     Min       1Q   Median       3Q      Max
-0.31846 -0.06473 -0.00914  0.06811  0.38879

Coefficients:
                           Estimate Std. Error t value
(Intercept)               5.4902552  0.2541738  21.600
L1.lnsales                0.3555113  0.0273145  13.015
lngoogle.ads              0.0419183  0.0033582  12.482
lnfacebook                0.0387449  0.0035752  10.837
lntiktok                  0.0469963  0.0042313  11.107
lnfacebook:lntiktok      -0.0005874  0.0004124  -1.424
lngoogle.ads:lntiktok    -0.0013487  0.0005479  -2.462
lngoogle.ads:lnfacebook -0.0011746  0.0005234  -2.244
                                    Pr(>|t|)
(Intercept)              <0.0000000000000002 ***
L1.lnsales               <0.0000000000000002 ***
lngoogle.ads             <0.0000000000000002 ***
lnfacebook               <0.0000000000000002 ***
lntiktok                 <0.0000000000000002 ***
lnfacebook:lntiktok                  0.1560
lngoogle.ads:lntiktok                0.0147 *
lngoogle.ads:lnfacebook              0.0260 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09949 on 191 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.8658,    Adjusted R-squared:  0.8608
F-statistic:   176 on 7 and 191 DF,  p-value: < 0.00000000000000022
```

<u>The interaction effect between TikTok and Facebook</u>

- Negative coefficient and not significant (b = –0.00059, p = .156), no clear evidence of either synergistic or antagonistic effect. The negative coefficient suggests a slight antagonistic effect but it is not significant, so we cannot affirm any interaction.

- Rationale by Channel Role:
    - TikTok: Highly viral, driven by sharing, and appealing to a younger audience.
    - Facebook: Enables precise targeting, reaches a more mature audience, generally less viral.
- Low levels of cross-channel interaction suggest these platforms function largely independently; using both simultaneously neither significantly boosts nor reduces overall sales.

The Interaction effect between TikTok and Google Ads

- Negative coefficient and significant p-value (b = –0.0013, p = .015*), which suggests an antagonistic effect, simultaneous spending on both channels reduces their incremental effectiveness.
- Rationale considering channel roles:
    - Google Ads: primarily awareness-building and intent capture (search ads, display).
    - TikTok: engagement-driven, viral content.
- Both channels drive attention, but together they may compete for the same audience's attention or budget, slightly reducing efficiency when both are ramped up simultaneously.

The Interaction effect between Facebook and Google Ads

- Negative coefficient and significant p-value (b = –0.0012, p = .026*), which suggests an antagonistic effect.
- Rationale considering channel roles:
    - Facebook: targeted reach and engagement.
    - Google Ads: awareness and intent.
- They may overlap in audience exposure; e.g., the same users see Google ads and Facebook ads, causing diminishing returns when both budgets are increased together.

Conclusion:

These findings imply that continuous significant investments in Google Ads and either Facebook or TikTok reduce marginal effectiveness.

Appendix

```
# Open data
data <- read.csv(file.choose())

# Retrieve variables
google.ads<-data$Google.Ads
facebook<-data$Facebook
tiktok <-data$TikTok
sales <- data$Sales

# Make the data time series data in R
# Frequency=52 for weekly time series
google.ads <- ts(google.ads,frequency = 52, start=c(2018,1))
facebook <- ts(facebook,frequency = 52, start=c(2018,1))
tiktok <- ts(tiktok,frequency = 52, start=c(2018,1))
sales <- ts(sales,frequency = 52, start=c(2018,1))


### Feel of the data
plot(sales, col="purple", main="Sales")
plot(tiktok, col="pink", main="Tiktok spending")
plot(facebook, col="blue", main="Facebook spending")
plot(google.ads, col="red", main="Google.Ads spending")

#Counting the shares of facebook, tiktok, and google ad
sum_tiktok<-sum(tiktok)
sum_facebook<-sum(facebook)
sum_googleads<-sum(google.ads)
total_spend<-sum_tiktok+sum_facebook+sum_googleads

tiktok_share<-sum_tiktok/total_spend
facebook_share<-sum_facebook/total_spend
googleads_share<-sum_googleads/total_spend

slices<-c(tiktok_share, facebook_share, googleads_share)
lbls<-c("Tiktok", "Facebook", "Google Ads")
pct<-round(slices*100)
lbls<-paste(lbls, pct)
lbls<-paste(lbls, "%", sep = "")
par(mfrow=c(1,1))
pie(slices, labels = lbls, col = rainbow(length(lbls)), main = "Ad Spending Share")


##Question 1 (2 points)

##Marketing Mix Modelling
#Linear regression
## Allow non scientific format
options(scipen=999)
### Assume linear relationship between ad spending and sales
##this does not work
regression1 <- lm(sales~google.ads+facebook+tiktok)
summary(regression1)
```

Dynamic effects

```
m <- 1   # one lag
#number of observations
n <- length(sales)
#Build Lag
L1.sales <- c(rep(NA,m), sales[1:(n-m)])
```

Allow non scientific format

```
options(scipen=999)
#this also doesnt work
regression2 <- lm(sales~L1.sales+google.ads+facebook+tiktok)
summary(regression2)
```

###Correct
Diminishing returns to scale

```
m <- 1   # one lag
#number of observations
n <- length(sales)
```

#Build Lag

```
L1.sales <- c(rep(NA,m), sales[1:(n-m)])
lngoogle.ads<-log(google.ads+1)
lnfacebook<-log(facebook+1)
lntiktok<-log(tiktok+1)
lnsales<-log(sales+1)

# We add ln to avoid undefined values of logarithm (log (0)  does not exist)
```

Lag for ln

```
L1.lnsales <- c(rep(NA,m), lnsales[1:(n-m)])
options(scipen=999)
regression3 <- lm(lnsales~L1.lnsales+lngoogle.ads+lnfacebook+lntiktok)
summary(regression3)
```

Model fit plot

```
fitted_sales<-ts(regression3$fitted.values, frequency = 52,
                 start=c(2018,1))
plot(lnsales, type="l", col="blue", main="Sales",lwd=2)
lines(fitted_sales, type="b", col="red")
legend("topleft", lty=1, col=c("blue", "red"),
       c("Logged Sales Data","Fitted"))
```

#Retrieve each model coefficient from regression output

```
# Recall - Log-log model is 'stored' in regression3

beta_adwords<-summary(regression3)$coefficients[3,1]
beta_facebook<-summary(regression3)$coefficients[4,1]
beta_tiktok<-summary(regression3)$coefficients[5,1]
```

These beta-s are the elasticities. We need to know how much one unit of ad changes one unit of sales. So, we need to convert them into unit effects

#Calculate the baseline (average) sales

```
average_sales<-mean(sales)

#Calculate the baseline (average) advertising spending for each media:
average_adwords<-mean(google.ads)
average_facebook<-mean(facebook)
average_tiktok <- mean(tiktok)

### Finally, calculate the unit effects:
theta_adwords<-beta_adwords*(average_sales/average_adwords)
theta_facebook<-beta_facebook*(average_sales/average_facebook)
theta_tiktok<-beta_tiktok*(average_sales/average_tiktok)


#How much sales we got thanks to tiktok, Adwords etc.?
# If we know one unit effect, we need to multiply with total spending.
sum_adwords<-sum(google.ads)
sum_facebook<-sum(facebook)
sum_tiktok<-sum(tiktok)

#Each media's contribution to sales
adwords_contribution<-theta_adwords*sum_adwords
facebook_contribution<-theta_facebook*sum_facebook
tiktok_contribution<-theta_tiktok*sum_tiktok


### Plot each media contribution in a nice bar plot
media_contribution<-c(adwords_contribution,facebook_contribution,
                      tiktok_contribution)

media_contribution=round(media_contribution, digits=0)
media_names<-c("AdWords","Facebook","Tiktok")
df<-data.frame(media_names,media_contribution)
head(df)

library(ggplot2)
barp_plot1<-ggplot(data=df,aes(x=media_names,y=media_contribution)) +
  geom_bar(stat="identity", color="black",
           fill=c("Red","Orange","Blue")) +
  geom_text(aes(label=media_contribution), vjust=-0.3, size=3.5)+
  labs(title="Contribution to Sales", x="Media", y="Contribution") +
  theme_minimal()

#bar_plot1

# change the order of the bars
barp_plot1 + scale_x_discrete(limits=c("Facebook","AdWords","Tiktok"))


# Question 2
#### Return on Marketing Investment
#Calculate the cost of each media
cost_adwords<-sum(google.ads)
cost_facebook<-sum(facebook)
cost_tiktok<-sum(tiktok)
cost_total<-cost_adwords+cost_facebook+cost_tiktok
```

```
cost<-c(cost_adwords,cost_facebook,cost_tiktok)
cost=round(cost, digits=0)

# Sales Contribution vs. Cost
library(ggplot2)
df3<-data.frame(sales_cost=rep(c("Sales","Cost"), each=3),
                media_names=rep(c("Adwords","Facebook","Tiktok"),2),
                values=c(media_contribution, cost))
head(df3)


barp_plot3<-ggplot(data = df3, aes(x = media_names,y = values,fill = sales_cost)) +
  geom_bar(stat = "identity", color = "black",
           position = position_dodge()) +
  labs(title = "Sales vs. Media Cost", x = "Media", y = "Sales and Cost") +
  theme_minimal()

bar_plot3<-barp_plot3 + scale_x_discrete(limits = c("Facebook", "Adwords", "Tiktok"))
bar_plot3


### Calculate ROI
# Calculate the traffic return for each media
roi_adwords=adwords_contribution/cost_adwords
roi_facebook=facebook_contribution/cost_facebook
roi_tiktok=tiktok_contribution/cost_tiktok

# TROMI plot input
roi<-c(roi_adwords, roi_facebook, roi_tiktok)

# Round off the numbers.
roi=round(roi, digits=2)
media_names<-c("Adwords", "Facebook","TikTok")
df4<-data.frame(media_names,roi)
head(df4)


# TROMI bar plot
library(ggplot2)

bar_plot4 <- ggplot(df4, aes(x = media_names, y = roi, fill = media_names)) +
  geom_bar(stat = "identity", color = "black") +
  scale_fill_manual(values = c("Adwords"="Red","Facebook"="Orange","TikTok"="Blue")) +
  geom_text(aes(label = roi), vjust = -0.3, size = 3.5) +
  labs(title = "Sales Return on Marketing Investment", x = "Media", y = "TROMI") +
  theme_minimal() +
  scale_x_discrete(limits=c("Facebook","Adwords","TikTok"))

bar_plot4


#Question3

##### Marketing Budget Allocation
```

#Actual Budget Spending

```
costshare_adwords<-cost_adwords/cost_total
costshare_facebook<-cost_facebook/cost_total
costshare_tiktok<-cost_tiktok/cost_total
```

#Input for the pie-chart

```
slices_actual<-c(costshare_adwords, costshare_facebook,costshare_tiktok )
lbls_actual<-c("Adwords", "Facebook", "TikTok")
pct_actual<-round(slices_actual*100)
lbls_actual<-paste(lbls_actual, pct_actual)
```

#Add data to labels

```
lbls_actual<-paste(lbls_actual, "%", sep= "")
```

Get the pie-chart

```
pie(slices_actual, labels = lbls_actual, col = rainbow(length(lbls_actual)), main = "Actual Ad Spending")
```

#The sum of all elasticities

```
beta_allmedia<-beta_adwords+beta_facebook+beta_tiktok
```

#Optimal resource allocation

```
optim_adwords<-beta_adwords/beta_allmedia
optim_facebook<-beta_facebook/beta_allmedia
optim_tiktok<-beta_tiktok/beta_allmedia
```

Pie-chart ingredients

```
optimal_spend<-c(optim_adwords,optim_facebook,optim_tiktok)
optimal_spend=round(optimal_spend, digits=2)
optimal_spend
slices_optim<-c(optim_adwords, optim_facebook,optim_tiktok)
lbls_optim<-c("Adwords", "Facebook", "Tiktok")
pct_optim<-round(slices_optim*100)
lbls_optim<-paste(lbls_optim, pct_optim)
```

#Paste variable names to data labels

```
lbls_optim<-paste(lbls_optim, "%", sep="") # add % sign to labels
```

Get the pie-chart

```
pie(slices_optim, labels=lbls_optim, col=rainbow(length(lbls_optim)), main="Optimal Budget Allocation" )
```

#Question4

Interactions

```
regression4 <- lm(lnsales~L1.lnsales+lngoogle.ads+lnfacebook+lntiktok
                  + lntiktok:lnfacebook+lntiktok:lngoogle.ads+lnfacebook:lngoogle.ads )
summary(regression4)
```